

Администрирование локальных сетей

Динамическая маршрутизация

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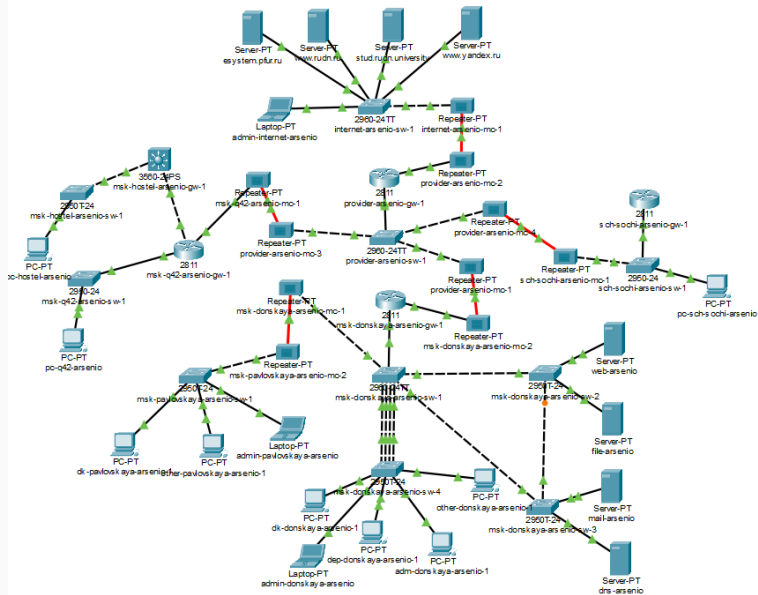
1 июля 2026

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Цели и задачи работы

Настроить динамическую маршрутизацию между территориями организации с использованием протокола OSPF.

Общая топология сети



Настройка OSPF

```
Password:
msk-donskaya-arsenio-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-arsenio-gw-1(config)#router ospf 1
msk-donskaya-arsenio-gw-1(config-router)#router-id 10.128.254.1
msk-donskaya-arsenio-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
msk-donskaya-arsenio-gw-1(config-router)#^Z
msk-donskaya-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-gw-1#
```

Рис. 2: Настройка OSPF на маршрутизаторе msk-donskaya-arsenio-gw-1

```
msk-q42-arsenio-gw-1>en
Password:
msk-q42-arsenio-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-arsenio-gw-1(config)#router ospf 1
msk-q42-arsenio-gw-1(config-router)#router-id 10.128.254.2
msk-q42-arsenio-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
msk-q42-arsenio-gw-1(config-router)#
01:10:28: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on FastEthernet0/1.5 from LOADING to
FULL, Loading Done

msk-q42-arsenio-gw-1(config-router)#^Z
msk-q42-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-arsenio-gw-1#wr mem
Building configuration...
[OK]
msk-q42-arsenio-gw-1#
```

Рис. 3: Настройка OSPF на маршрутизаторе msk-q42-arsenio-gw-1


```
msk-hostel-arsenio-gw-1>en
Password:
msk-hostel-arsenio-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-arsenio-gw-1(config)#router ospf 1
msk-hostel-arsenio-gw-1(config-router)#router-id 10.128.254.3
msk-hostel-arsenio-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
msk-hostel-arsenio-gw-1(config-router)#
01:11:08: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.2 on Vlan202 from LOADING to FULL,
Loading Done

msk-hostel-arsenio-gw-1(config-router)#^Z
msk-hostel-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-arsenio-gw-1#wr mem
Building configuration...
[OK]
msk-hostel-arsenio-gw-1#
```

Рис. 4: Настройка OSPF на маршрутизаторе msk-hostel-arsenio-gw-1

```
sch-sochi-arsenio-gw-1>en
Password:
sch-sochi-arsenio-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-arsenio-gw-1(config)#router ospf 1
sch-sochi-arsenio-gw-1(config-router)#router-id 10.128.254.4
sch-sochi-arsenio-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
sch-sochi-arsenio-gw-1(config-router)#
01:09:13: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on FastEthernet0/0.6 from LOADING to
FULL, Loading Done

sch-sochi-arsenio-gw-1(config-router)#^Z
sch-sochi-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-arsenio-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-arsenio-gw-1#
```

Рис. 5: Настройка OSPF на маршрутизаторе sch-sochi-arsenio-gw-1

Прямая связь q42-sochi

Создание VLAN 7 у провайдера

```
provider-arsenio-sw-1>en
Password:
provider-arsenio-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-arsenio-sw-1(config)#
provider-arsenio-sw-1(config)#vlan 7
provider-arsenio-sw-1(config-vlan)#name q42-sochi
provider-arsenio-sw-1(config-vlan)#int vlan 7
provider-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

provider-arsenio-sw-1(config-if)#no sh
provider-arsenio-sw-1(config-if)#^Z
provider-arsenio-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-arsenio-sw-1#wr mem
Building configuration...
[OK]
provider-arsenio-sw-1#
```

Рис. 6: Создание VLAN 7 q42-sochi на коммутаторе provider-arsenio-sw-1

```
msk-q42-arsenio-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-arsenio-gw-1(config)#int f0/1.7
msk-q42-arsenio-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.7, changed state to up

msk-q42-arsenio-gw-1(config-subif)#enc dot1q 7
msk-q42-arsenio-gw-1(config-subif)#ip addr 10.128.255.9 255.255.255.252
msk-q42-arsenio-gw-1(config-subif)#descr sochi
msk-q42-arsenio-gw-1(config-subif)#^Z
msk-q42-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-arsenio-gw-1#wr mem
Building configuration...
[OK]
msk-q42-arsenio-gw-1#
```

Рис. 7: Настройка подинтерфейса f0/1.7 на маршрутизаторе msk-q42-arsenio-gw-1

```
sch-sochi-arsenio-sw-1>en
Password:
sch-sochi-arsenio-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-arsenio-sw-1(config)#vlan 7
sch-sochi-arsenio-sw-1(config-vlan)#name q42-sochi
sch-sochi-arsenio-sw-1(config-vlan)#int vlan 7
sch-sochi-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

sch-sochi-arsenio-sw-1(config-if)#no sh
sch-sochi-arsenio-sw-1(config-if)#^Z
sch-sochi-arsenio-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-arsenio-sw-1#wr mem
Building configuration...
[OK]
sch-sochi-arsenio-sw-1#
```

Рис. 8: Создание VLAN 7 q42-sochi на коммутаторе sch-sochi-arsenio-sw-1

Подинтерфейс на sch-sochi-gw-1

```
sch-sochi-arsenio-gw-1#
sch-sochi-arsenio-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-arsenio-gw-1(config)#int f0/0.7
sch-sochi-arsenio-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.7, changed state to up

sch-sochi-arsenio-gw-1(config-subif)#enc dot1q 7
sch-sochi-arsenio-gw-1(config-subif)#ip addr 10.128.255.10 255.255.255.252
sch-sochi-arsenio-gw-1(config-subif)#descr q 42
sch-sochi-arsenio-gw-1(config-subif)#
01:12:35: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.2 on FastEthernet0/0.7 from LOADING to
FULL, Loading Done

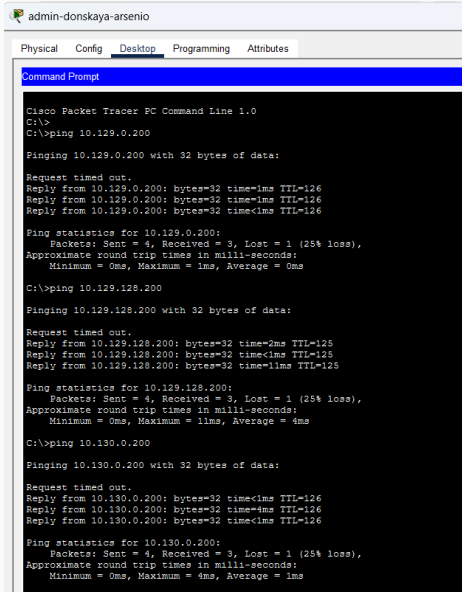
sch-sochi-arsenio-gw-1(config-subif)#^Z
sch-sochi-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-arsenio-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-arsenio-gw-1#
```

Рис. 9: Настройка подинтерфейса f0/0.7 на маршрутизаторе sch-sochi-arsenio-gw-1

Проверка связи

Проверка ping



```
admin-donskaya-arsenio
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 10.129.0.200

Pinging 10.129.0.200 with 32 bytes of data:

Request timed out.
Reply from 10.129.0.200: bytes=32 time=1ms TTL=126
Reply from 10.129.0.200: bytes=32 time=1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.129.0.200:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.129.128.200

Pinging 10.129.128.200 with 32 bytes of data:

Request timed out.
Reply from 10.129.128.200: bytes=32 time=2ms TTL=125
Reply from 10.129.128.200: bytes=32 time<1ms TTL=125
Reply from 10.129.128.200: bytes=32 time=11ms TTL=125

Ping statistics for 10.129.128.200:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 4ms

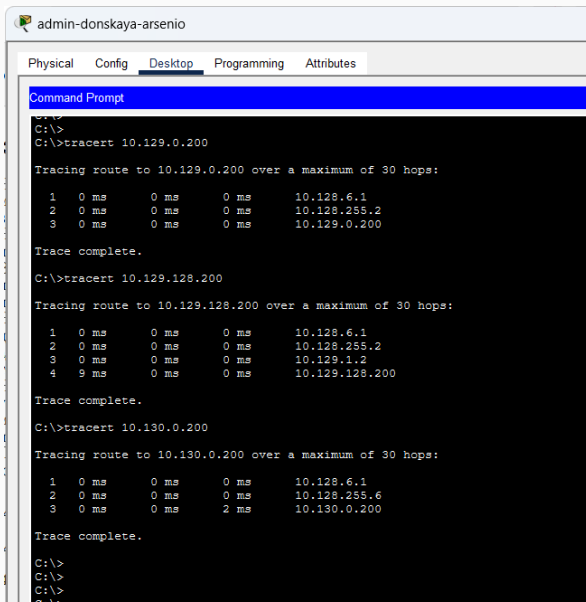
C:\>ping 10.130.0.200

Pinging 10.130.0.200 with 32 bytes of data:

Request timed out.
Reply from 10.130.0.200: bytes=32 time<1ms TTL=126
Reply from 10.130.0.200: bytes=32 time=4ms TTL=126
Reply from 10.130.0.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

Проверка tracert



The screenshot shows a Windows desktop with a taskbar at the bottom. The active window is titled "admin-donskaya-arsenio" and has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is selected, and a "Command Prompt" window is open on top of it. The Command Prompt shows the execution of three tracert commands. The first command is "tracert 10.129.0.200", which shows a 3-hop route through 10.128.6.1 and 10.128.255.2. The second command is "tracert 10.129.128.200", which shows a 4-hop route through 10.128.6.1, 10.128.255.2, and 10.129.1.2. The third command is "tracert 10.130.0.200", which shows a 3-hop route through 10.128.6.1 and 10.128.255.6. Each trace is followed by the message "Trace complete."

```
C:\>
C:\>
C:\>tracert 10.129.0.200

Tracing route to 10.129.0.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.128.6.1
  2  0 ms    0 ms    0 ms    10.128.255.2
  3  0 ms    0 ms    0 ms    10.129.0.200

Trace complete.

C:\>tracert 10.129.128.200

Tracing route to 10.129.128.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.128.6.1
  2  0 ms    0 ms    0 ms    10.128.255.2
  3  0 ms    0 ms    0 ms    10.129.1.2
  4  9 ms    0 ms    0 ms    10.129.128.200

Trace complete.

C:\>tracert 10.130.0.200

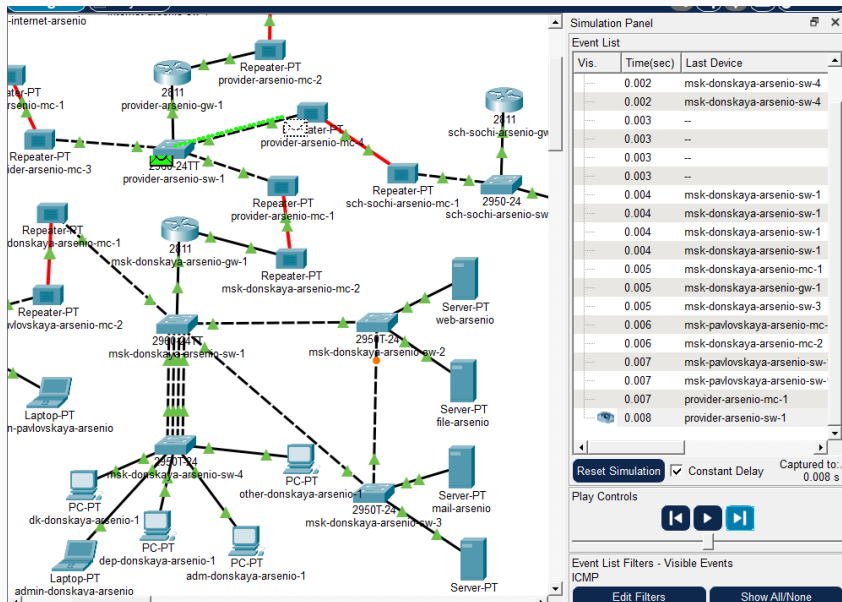
Tracing route to 10.130.0.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.128.6.1
  2  0 ms    0 ms    0 ms    10.128.255.6
  3  0 ms    0 ms    2 ms    10.130.0.200

Trace complete.

C:\>
C:\>
C:\>
C:\>
```

ICMP в режиме Simulation



Проверка отказоустойчивости

```
provider-arsenio-sw-1>en
Password:
provider-arsenio-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-arsenio-sw-1(config)#int vlan 6
provider-arsenio-sw-1(config-if)#sh

provider-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to down

provider-arsenio-sw-1(config-if)#no int vlan 6
provider-arsenio-sw-1(config)#no vlan 6
provider-arsenio-sw-1(config)#
```

Рис. 13: Временное отключение интерфейса Vlan6 на коммутаторе provider-arsenio-sw-1

Маршрут после отключения VLAN 6

```
*C
C:\>ping 10.130.0.200

Pinging 10.130.0.200 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.130.0.200: bytes=32 time<1ms TTL=125
Reply from 10.130.0.200: bytes=32 time<1ms TTL=125

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
C:\>
C:\>
C:\>tracert 10.130.0.200

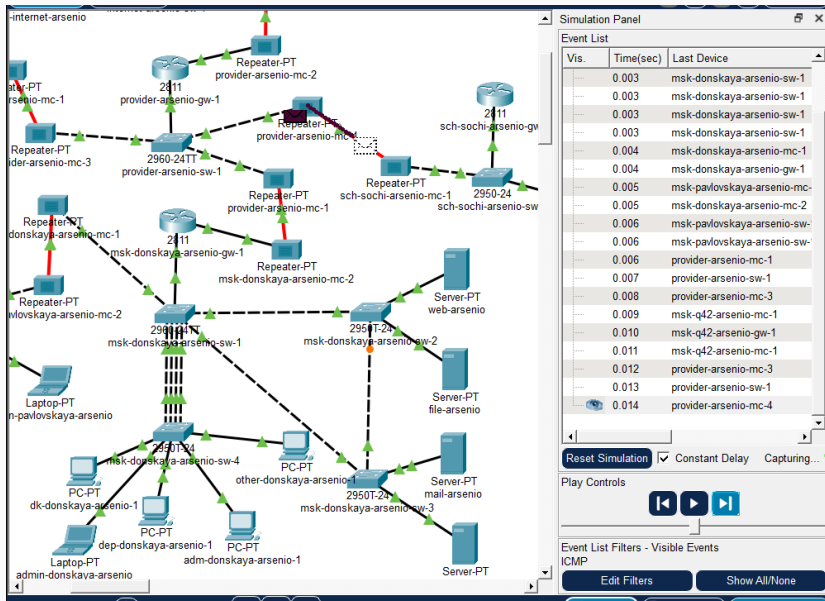
Tracing route to 10.130.0.200 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.128.6.1
  1  0 ms    0 ms    0 ms    10.128.255.2
  2  0 ms    0 ms    2 ms    10.128.255.10
  3  1 ms    0 ms    0 ms    10.130.0.200

Trace complete.

C:\>
```

ICMP после отключения VLAN 6



```
provider-arsenio-sw-1(config)#
provider-arsenio-sw-1(config)#vlan 6
provider-arsenio-sw-1(config-vlan)#name sochi
provider-arsenio-sw-1(config-vlan)#int vlan 6
provider-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

provider-arsenio-sw-1(config-if)#no shut
provider-arsenio-sw-1(config-if)#^Z
provider-arsenio-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-arsenio-sw-1#wr mem
Building configuration...
[OK]
provider-arsenio-sw-1#
```

Рис. 16: Восстановление интерфейса Vlan6 на коммутаторе provider-arsenio-sw-1


```
C:\>
C:\>ping 10.130.0.200

Pinging 10.130.0.200 with 32 bytes of data:

Reply from 10.130.0.200: bytes=32 time=5ms TTL=125
Reply from 10.130.0.200: bytes=32 time=12ms TTL=125
Reply from 10.130.0.200: bytes=32 time<1ms TTL=125
Reply from 10.130.0.200: bytes=32 time=1ms TTL=125

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 4ms

C:\>tracert 10.130.0.200

Tracing route to 10.130.0.200 over a maximum of 30 hops:

  0  5 ms    2 ms     0 ms    10.128.6.1
  1  0 ms     0 ms     1 ms    10.128.255.6
  2  0 ms     0 ms     2 ms    10.130.0.200

Trace complete.

C:\>
```

Проверка OSPF и маршрутов

OSPF на msk-donskaya-gw-1

```
msk-donskaya-arsenio-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.1
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 8
    Area has no authentication
    SPF algorithm executed 6 times
    Area ranges are
    Number of LSA 8. Checksum Sum 0x045032
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
```

```
msk-donskaya-arsenio-gw-1#sh ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.128.254.2	1	FULL/DR	00:00:38	10.128.255.2	FastEthernet0/1.5
10.128.254.4	1	FULL/DR	00:00:38	10.128.255.6	FastEthernet0/1.6

```
msk-donskaya-arsenio-gw-1#
```

Таблица маршрутизации msk-donskaya-gw-1

```
msk-donskaya-arsenio-gw-1#
msk-donskaya-arsenio-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 24 subnets, 4 masks
C       10.128.0.0/24 is directly connected, FastEthernet0/0.3
L       10.128.0.1/32 is directly connected, FastEthernet0/0.3
C       10.128.1.0/24 is directly connected, FastEthernet0/0.2
L       10.128.1.1/32 is directly connected, FastEthernet0/0.2
C       10.128.3.0/24 is directly connected, FastEthernet0/0.101
L       10.128.3.1/32 is directly connected, FastEthernet0/0.101
C       10.128.4.0/24 is directly connected, FastEthernet0/0.102
L       10.128.4.1/32 is directly connected, FastEthernet0/0.102
C       10.128.5.0/24 is directly connected, FastEthernet0/0.103
L       10.128.5.1/32 is directly connected, FastEthernet0/0.103
C       10.128.6.0/24 is directly connected, FastEthernet0/0.104
L       10.128.6.1/32 is directly connected, FastEthernet0/0.104
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.1/32 is directly connected, FastEthernet0/1.5
C       10.128.255.4/30 is directly connected, FastEthernet0/1.6
L       10.128.255.5/32 is directly connected, FastEthernet0/1.6
O       10.128.255.8/30 [110/2] via 10.128.255.2, 00:00:49, FastEthernet0/1.5
        [110/2] via 10.128.255.6, 00:00:49, FastEthernet0/1.6
S       10.129.0.0/16 [1/0] via 10.128.255.2
O       10.129.0.0/24 [110/2] via 10.128.255.2, 00:05:19, FastEthernet0/1.5
O       10.129.1.0/24 [110/2] via 10.128.255.2, 00:05:19, FastEthernet0/1.5
O       10.129.128.0/24 [110/3] via 10.128.255.2, 00:05:19, FastEthernet0/1.5
S       10.130.0.0/16 [1/0] via 10.128.255.6
O       10.130.0.0/24 [110/2] via 10.128.255.6, 00:00:49, FastEthernet0/1.6
O       10.130.1.0/24 [110/2] via 10.128.255.6, 00:00:49, FastEthernet0/1.6
    198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/28 is directly connected, FastEthernet0/1.4
L       198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*    0.0.0.0/0 [1/0] via 198.51.100.1

msk-donskaya-arsenio-gw-1#
```

```
msk-q42-arsenio-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.2
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 4
    Area has no authentication
    SPF algorithm executed 7 times
    Area ranges are
    Number of LSA 8. Checksum Sum 0x045032
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
```

```
msk-q42-arsenio-gw-1#sh ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.128.254.1	1	FULL/BDR	00:00:33	10.128.255.1	FastEthernet0/1.5
10.128.254.4	1	FULL/DR	00:00:33	10.128.255.10	FastEthernet0/1.7
10.128.254.3	1	FULL/DR	00:00:33	10.129.1.2	FastEthernet1/0.202

```
msk-q42-arsenio-gw-1#
```

Таблица маршрутизации msk-q42-gw-1

```
msk-q42-arsenio-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.1 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 19 subnets, 4 masks
O    10.128.0.0/24 [110/2] via 10.128.255.1, 00:05:46, FastEthernet0/1.5
O    10.128.1.0/24 [110/2] via 10.128.255.1, 00:05:46, FastEthernet0/1.5
O    10.128.3.0/24 [110/2] via 10.128.255.1, 00:05:46, FastEthernet0/1.5
O    10.128.4.0/24 [110/2] via 10.128.255.1, 00:05:46, FastEthernet0/1.5
O    10.128.5.0/24 [110/2] via 10.128.255.1, 00:05:46, FastEthernet0/1.5
O    10.128.6.0/24 [110/2] via 10.128.255.1, 00:05:46, FastEthernet0/1.5
C    10.128.255.0/30 is directly connected, FastEthernet0/1.5
L    10.128.255.2/32 is directly connected, FastEthernet0/1.5
O    10.128.255.4/30 [110/2] via 10.128.255.1, 00:01:26, FastEthernet0/1.5
      [110/2] via 10.128.255.10, 00:01:26, FastEthernet0/1.7
C    10.128.255.8/30 is directly connected, FastEthernet0/1.7
L    10.128.255.9/32 is directly connected, FastEthernet0/1.7
C    10.129.0.0/24 is directly connected, FastEthernet0/0.201
L    10.129.0.1/32 is directly connected, FastEthernet0/0.201
C    10.129.1.0/24 is directly connected, FastEthernet1/0.202
L    10.129.1.1/32 is directly connected, FastEthernet1/0.202
S    10.129.128.0/17 [1/0] via 10.129.1.2
O    10.129.128.0/24 [110/2] via 10.129.1.2, 00:06:06, FastEthernet1/0.202
O    10.130.0.0/24 [110/2] via 10.128.255.10, 00:05:56, FastEthernet0/1.7
O    10.130.1.0/24 [110/2] via 10.128.255.10, 00:05:56, FastEthernet0/1.7
S*   0.0.0.0/0 [1/0] via 10.128.255.1

msk-q42-arsenio-gw-1#
```

```
msk-hostel-arsenio-gw-1#
```

```
msk-hostel-arsenio-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.3
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 2
    Area has no authentication
    SPF algorithm executed 6 times
    Area ranges are
    Number of LSA 8. Checksum Sum 0x045032
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
```

```
msk-hostel-arsenio-gw-1#sh ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.128.254.2	1	FULL/BDR	00:00:33	10.129.1.1	Vlan202

```
msk-hostel-arsenio-gw-1#
msk-hostel-arsenio-gw-1#
```

Таблица маршрутизации msk-hostel-gw-1

```
msk-hostel-arsenio-gw-1#sh ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.128.254.2	1	FULL/BDR	00:00:33	10.129.1.1	Vlan202

```
msk-hostel-arsenio-gw-1#
```

```
msk-hostel-gw-1#
```

```
msk-hostel-arsenio-gw-1#sh ip rou
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is 10.129.1.1 to network 0.0.0.0
```

```
10.0.0.0/8 is variably subnetted, 14 subnets, 2 masks
```

```
O    10.128.0.0/24 [110/3] via 10.129.1.1, 00:06:57, Vlan202
O    10.128.1.0/24 [110/3] via 10.129.1.1, 00:06:57, Vlan202
O    10.128.3.0/24 [110/3] via 10.129.1.1, 00:06:57, Vlan202
O    10.128.4.0/24 [110/3] via 10.129.1.1, 00:06:57, Vlan202
O    10.128.5.0/24 [110/3] via 10.129.1.1, 00:06:57, Vlan202
O    10.128.6.0/24 [110/3] via 10.129.1.1, 00:06:57, Vlan202
O    10.128.255.0/30 [110/2] via 10.129.1.1, 00:07:12, Vlan202
O    10.128.255.4/30 [110/3] via 10.129.1.1, 00:02:32, Vlan202
O    10.128.255.8/30 [110/2] via 10.129.1.1, 00:06:57, Vlan202
O    10.129.0.0/24 [110/2] via 10.129.1.1, 00:07:12, Vlan202
C    10.129.1.0/24 is directly connected, Vlan202
C    10.129.128.0/24 is directly connected, Vlan301
O    10.130.0.0/24 [110/3] via 10.129.1.1, 00:06:57, Vlan202
O    10.130.1.0/24 [110/3] via 10.129.1.1, 00:06:57, Vlan202
S*   0.0.0.0/0 [1/0] via 10.129.1.1
```

```
msk-hostel-arsenio-gw-1#
```



```
sch-sochi-arsenio-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.4
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 4
    Area has no authentication
    SPF algorithm executed 6 times
    Area ranges are
    Number of LSA 8. Checksum Sum 0x045032
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
```

```
sch-sochi-arsenio-gw-1#sh ip ospf nei
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.128.254.1	1	FULL/BDR	00:00:36	10.128.255.5	FastEthernet0/0.6
10.128.254.2	1	FULL/BDR	00:00:36	10.128.255.9	FastEthernet0/0.7

```
sch-sochi-arsenio-gw-1#
```

Таблица маршрутизации sch-sochi-gw-1

```
sch-sochi-arsenio-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.5 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 18 subnets, 3 masks
O       10.128.0.0/24 [110/2] via 10.128.255.5, 00:03:08, FastEthernet0/0.6
O       10.128.1.0/24 [110/2] via 10.128.255.5, 00:03:08, FastEthernet0/0.6
O       10.128.3.0/24 [110/2] via 10.128.255.5, 00:03:08, FastEthernet0/0.6
O       10.128.4.0/24 [110/2] via 10.128.255.5, 00:03:08, FastEthernet0/0.6
O       10.128.5.0/24 [110/2] via 10.128.255.5, 00:03:08, FastEthernet0/0.6
O       10.128.6.0/24 [110/2] via 10.128.255.5, 00:03:08, FastEthernet0/0.6
O       10.128.255.0/30 [110/2] via 10.128.255.5, 00:03:08, FastEthernet0/0.6
           [110/2] via 10.128.255.9, 00:03:08, FastEthernet0/0.7
C       10.128.255.4/30 is directly connected, FastEthernet0/0.6
L       10.128.255.6/32 is directly connected, FastEthernet0/0.6
C       10.128.255.8/30 is directly connected, FastEthernet0/0.7
L       10.128.255.10/32 is directly connected, FastEthernet0/0.7
O       10.129.0.0/24 [110/2] via 10.128.255.9, 00:07:43, FastEthernet0/0.7
O       10.129.1.0/24 [110/2] via 10.128.255.9, 00:07:43, FastEthernet0/0.7
O       10.129.128.0/24 [110/3] via 10.128.255.9, 00:07:43, FastEthernet0/0.7
C       10.130.0.0/24 is directly connected, FastEthernet0/0.401
L       10.130.0.1/32 is directly connected, FastEthernet0/0.401
C       10.130.1.0/24 is directly connected, FastEthernet0/0.402
L       10.130.1.1/32 is directly connected, FastEthernet0/0.402
S*     0.0.0.0/0 [1/0] via 10.128.255.5

sch-sochi-arsenio-gw-1#
```

Выводы

В ходе лабораторной работы была настроена динамическая маршрутизация по протоколу OSPF между территориями организации.

Была создана прямая связь между кварталом 42 в Москве и филиалом в г. Сочи через VLAN 7, проверена доступность удалённых сетей, а также исследовано изменение маршрута ICMP-пакета при отключении и восстановлении VLAN 6.